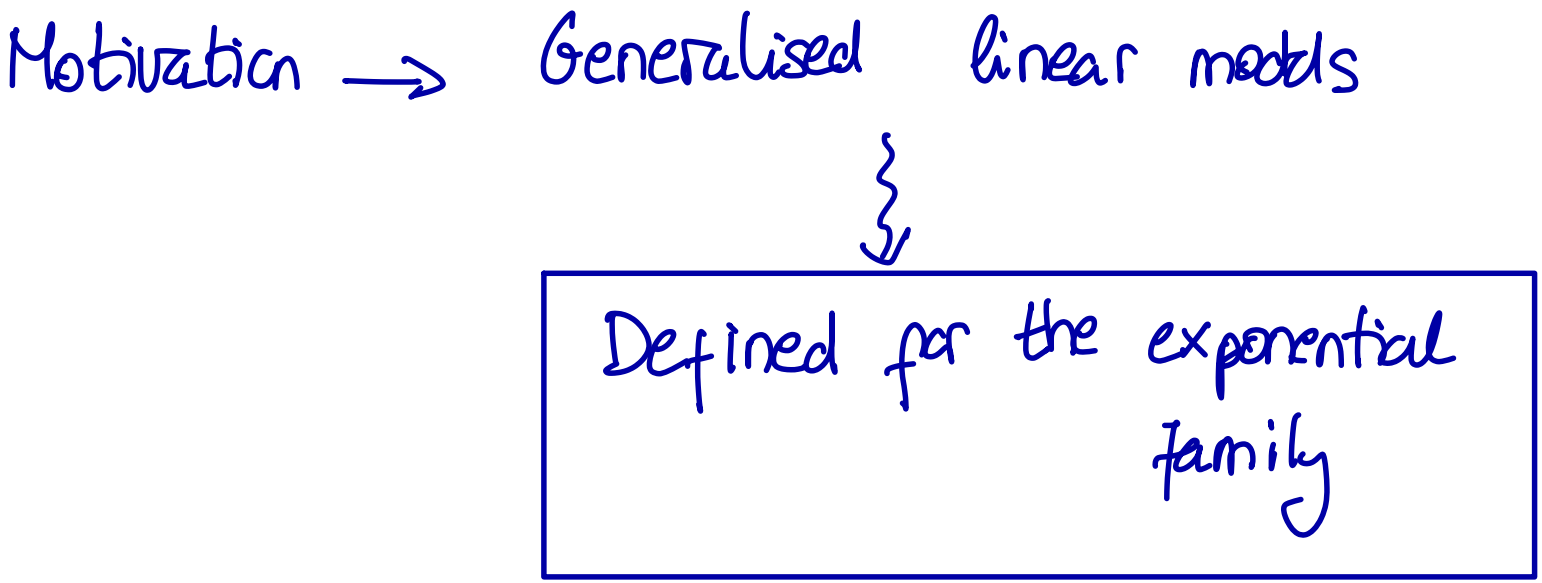
Assignment available → Friday 14 at 5pm

due date → Friday 21 at 5pm

- Submission → print to pdf html report
  - Latex or hand written equations.
- Special consideration → Please don't email the teaching team.
  - New central system

Exponential families



Learning goals

- Be able to show whether a given distribution is an exponential family.
- Be able to compute mean and variance of random variables belonging to exponential families.
- Understand the interpretation of the variance function and be able to compute the variance function for exponential families.

Exponential families

Y comes from an exponential family if it has density/mass function of the form

$$f(y;\theta,\phi) = \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right]$$

- θ is the *canonical parameter* (captures location)
- ϕ is the *dispersion parameter* (captures scale)

↦ Linear function of Y and $b(\theta)$ depends only on θ

↦ Any function of Y and ϕ

Example: normal

$Y \sim N(\mu, \sigma^2)$

$$\begin{aligned} f(y) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(y-\mu)^2}{\sigma^2}} \\ &= \exp\left[\frac{y\mu - \mu^2/2}{\sigma^2} - \frac{1}{2}\left(\frac{y^2}{\sigma^2} + \log(2\pi\sigma^2)\right)\right] \\ &= \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right] \end{aligned}$$

where $\theta = \mu$, $\phi = \sigma^2$, and

$$\begin{aligned} b(\theta) &= \theta^2/2 \\ a(\phi) &= \phi \\ c(y, \phi) &= -\frac{1}{2}\left(\frac{y^2}{\phi} + \log(2\pi\phi)\right) \end{aligned}$$

$$Y \sim N(\mu, \sigma^2) \Rightarrow f(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2}\frac{(y-\mu)^2}{\sigma^2}\right]$$

$$f(y;\theta,\phi) = \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right] \Rightarrow$$

$$f(y) = \exp\left\{\log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{(y-\mu)^2}{2\sigma^2}\right\}$$

$$= \exp\left\{\left[\log(2\pi\sigma^2)^{-1/2}\right] + \left[-\frac{y^2+\mu^2-2\mu y}{2\sigma^2}\right]\right\}$$

$$= \exp\left\{\frac{2y\mu}{2\sigma^2} - \frac{\mu^2}{2\sigma^2} + \left(\frac{-y^2}{2\sigma^2}\right) + \log(2\pi\sigma^2)^{-1/2}\right\}$$

$$= \exp\left\{\frac{y\mu - \frac{\mu^2}{2}}{\sigma^2} + \left(\frac{-y^2}{2\sigma^2} + \log(2\pi\sigma^2)^{-1/2}\right)\right\} \leftarrow$$

$$b(\theta) = \frac{\theta^2}{2} \quad \text{with} \quad \theta = \mu \quad \text{(please write it in terms of } \theta\text{)}$$

$$a(\phi) = \phi \quad \text{with} \quad \phi = \sigma^2$$

$$c(y, \phi) = -\frac{1}{2}\left(\frac{y^2}{\phi} + \log 2\pi\phi\right)$$

The parametrization is not unique. For example we could consider $\phi = \sigma$ and that we will also be correct.

Example: Poisson

$Y \sim \text{pois}(\lambda)$

$$\begin{aligned} f(y) &= e^{-\lambda}\lambda^y/y! \text{ for } y = 0, 1, 2, \ldots \\ &= \exp\left[y \log \lambda - \lambda - \log y!\right] \\ &= \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right] \end{aligned}$$

where $\theta = \log \lambda$, $\phi = 1$, and

$$\begin{aligned} b(\theta) &= e^{\theta} \\ a(\phi) &= \phi \\ c(y, \phi) &= -\log y! \end{aligned}$$

$Y \sim \text{Poisson}(\lambda) \Rightarrow f(y) = \dfrac{e^{-\lambda}\lambda^{y}}{y!}$

$$f(y;\theta,\phi) = \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right]$$

$$f(y) = \exp\{-\lambda + y\log\lambda - \log y!\}$$

$$= \exp\left\{\frac{y\log\lambda - \lambda}{1} + (-\log y!)\right\}$$

$\lambda = \theta \Rightarrow b(\theta) = \lambda = e^{\theta}$

$\phi = 1 \Rightarrow a(\phi) = \phi$

$c(y,\phi) = -\log y!$

Example: binomial

$Y \sim \text{bin}(m, p)$ for known m (not a parameter)

$$f(y) = \binom{m}{y} p^y (1-p)^{m-y} \text{ for } y = 0, 1, \ldots, m$$

Lab problem in the week 4.

Other examples of exponential families are the gamma and the inverse Gaussian.

Exponential family: mean and variance

Lemma If Y is from an exponential family then

$$\begin{aligned} \mathbb{E}\, Y &= b'(\theta) \\ \operatorname{Var} Y &= b''(\theta) a(\phi) \end{aligned}$$

[Proof] Exercise.

Let's use the lemma to study the mean and variance:

- Normal distribution
- Poisson distribution

Lemma If Y is from an exponential family then

$$\begin{aligned} \mathbb{E}\, Y &= b'(\theta) \\ \operatorname{Var} Y &= b''(\theta) a(\phi) \end{aligned}$$

$Y \sim N(\mu, \sigma^2)$

$$\begin{aligned} f(y) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(y-\mu)^2}{\sigma^2}} \\ &= \exp\left[\frac{y\mu - \mu^2/2}{\sigma^2} - \frac{1}{2}\left(\frac{y^2}{\sigma^2} + \log(2\pi\sigma^2)\right)\right] \\ &= \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right] \end{aligned}$$

where $\theta = \mu$, $\phi = \sigma^2$, and

$$\begin{aligned} b(\theta) &= \theta^2/2 \\ a(\phi) &= \phi \\ c(y,\phi) &= -\frac{1}{2}\left(\frac{y^2}{\phi} + \log(2\pi\phi)\right) \end{aligned}$$

$\mathbb{E}(Y) = \frac{d}{d\theta}\left(\frac{\theta^2}{2}\right) = \theta \rightarrow \theta = \mu$

$\operatorname{Var}(Y) = 1 \cdot \phi = \phi \rightarrow \phi = \sigma^2$

Lemma If Y is from an exponential family then

$$\begin{aligned} \mathbb{E}\, Y &= b'(\theta) \\ \operatorname{Var} Y &= b''(\theta) a(\phi) \end{aligned}$$

Example: Poisson

$Y \sim \text{pois}(\lambda)$

$$\begin{aligned} f(y) &= e^{-\lambda}\lambda^y / y! \text{ for } y = 0, 1, 2, \ldots \\ &= \exp\left[y \log \lambda - \lambda - \log y!\right] \\ &= \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right] \end{aligned}$$

where $\theta = \log \lambda$, $\phi = 1$, and

$$\begin{aligned} b(\theta) &= e^{\theta} \\ a(\phi) &= \phi \\ c(y, \phi) &= -\log y! \end{aligned}$$

$$E(Y) = \frac{d}{d\theta} e^{\theta} = e^{\theta} = e^{\log \lambda} = \lambda$$

$$\text{Var}(Y) = \frac{d^2 e^{\theta}}{d\theta} \times \phi = e^{\theta} = \lambda$$

$$\phi = 1$$

Different parametrizations lead to the same $E(y)$ $\text{Var}(y)$!

Exponential family: variance function

$Y \sim$ exponential family $\Rightarrow$

Let $\mu = \mathbb{E}\, Y$ and write

$$\operatorname{Var} Y = v(\mu)a(\phi)$$

(so $v = b'' \circ (b')^{-1}$). v is called the *variance function*

Examples:

normal	$v(\mu) = 1$
Poisson	$v(\mu) = \mu$
binomial	$v(\mu) = \mu(1 - \mu/m)$

Intuition:

The mean and variance are not always independent!

Variance → how the spread of Y changes with μ.

▶ $v(\mu)$ is a function of μ

Let $\mu = \mathbb{E}\, Y$ and write

$$\operatorname{Var} Y = v(\mu)a(\phi)$$

(so $v = b'' \circ (b')^{-1}$). v is called the *variance function*

$\mu = \mathbb{E}(Y) = b'(\theta) \rightarrow (b')^{-1}(\mu) = \theta$

$\operatorname{Var}(Y) = b''(\theta)\, a(\phi)$

$= b''\left((b')^{-1}(\mu)\right) a(\phi)$

$= v(\mu) \cdot a(\phi)$

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